CS 215: Task 02: Due: 15-Aug; time as announced on Piazza. No extensions.

- Your answers should be in your handwriting. Try to be neat and clear. If an LLM can read your handwriting, we might be able to.
- Please write the honor code statement only if it is true. If you used any source (person or thing) explicitly state it. This is an individual assignment.

Teaching-team notes.

- When we say, 'derive' or 'show', simply put 2-3 important steps in the derivation in the space provided. This is practice for your midsemester exam, where we may not be able to give extra space. You have to decide what is important. For the homework, please attach extra paper for your records and for you to scan in the upload.

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0. Please write the honor code in your handwriting, with the understanding that it is true.

Solution:

1. $U \sim \text{Uniform}(0,1)$ and $a < b$, define
$$U' = a + (b-a)U.$$

(a) Show that U' is uniformly distributed on (a,b) by working with the cumulative distribution function of U'.

Teaching-team notes. Do not use any shortcut such as the Leibniz rule discussed in class.

Solution: For $a < x < b$,
$$\begin{aligned} P(U' \le x) &= P\left(a + (b-a)U \le x\right) \\ &= P\left(U \le \frac{x-a}{b-a}\right) \\ &= \frac{x-a}{b-a}. \end{aligned}$$

Therefore, the cumulative distribution function of U' is
$$F_{U'}(x) = \begin{cases} 0, & x \le a, \\ \dfrac{x-a}{b-a}, & a < x < b, \\ 1, & x \ge b. \end{cases}$$

This is precisely the CDF of a $\text{Uniform}(a,b)$ random variable. Hence,
$$\boxed{U' \sim \text{Uniform}(a,b).}$$

(b) Derive the moment-generating function of U'.

Solution: Since $U' \sim \text{Uniform}(a, b)$, its density is

$$f_{U'}(x) = \frac{1}{b-a}, \qquad a < x < b.$$

Hence, for $t \neq 0$,

$$\begin{aligned} M_{U'}(t) &= E(e^{tU'}) \\ &= \frac{1}{b-a} \int_a^b e^{tx}\, dx \\ &= \frac{1}{b-a} \left[\frac{e^{tx}}{t} \right]_a^b \\ &= \frac{e^{bt} - e^{at}}{(b-a)t}. \end{aligned}$$

At $t = 0$, by continuity,

$$M_{U'}(0) = 1.$$

Therefore,

$$\boxed{M_{U'}(t) = \begin{cases} \dfrac{e^{bt} - e^{at}}{(b-a)t}, & t \neq 0, \\ 1, & t = 0. \end{cases}}$$

(c) Use the MGF to compute $E(U')$ and $\text{Var}(U')$.

Solution: Using the Taylor expansion of the MGF around $t = 0$,

$$M_{U'}(t) = 1 + E(U')t + \frac{E((U')^2)}{2!}t^2 + \cdots.$$

Expanding the MGF,

$$M_{U'}(t) = \frac{e^{bt} - e^{at}}{(b-a)t} = 1 + \frac{a+b}{2}t + \frac{a^2 + ab + b^2}{6}t^2 + \cdots.$$

Therefore,

$$E(U') = M'_{U'}(0) = \frac{a+b}{2},$$

and

$$E((U')^2) = M''_{U'}(0) = \frac{a^2 + ab + b^2}{3}.$$

Hence,

$$\begin{aligned} \text{Var}(U') &= E((U')^2) - [E(U')]^2 \\ &= \frac{a^2 + ab + b^2}{3} - \left(\frac{a+b}{2} \right)^2 \\ &= \frac{(b-a)^2}{12}. \end{aligned}$$

Thus,

$$\boxed{E(U') = \frac{a+b}{2}}, \qquad \boxed{\text{Var}(U') = \frac{(b-a)^2}{12}}.$$

2. $X \sim \text{Hypergeometric}(M, N, K)$, thus,

$$P(X = x \mid M, N, K) = \frac{\binom{M}{x}\binom{N-M}{K-x}}{\binom{N}{K}}, \qquad x = 0, 1, \ldots, K.$$

In class we discussed the relationship of the expectation and the variance of X with the binomial distribution. Now, show, using Stirling's approximation, viz., $n! \approx \sqrt{2\pi n}\left(\frac{n}{e}\right)^n$, that if $N \to \infty$, $M \to \infty$, and

$$\frac{M}{N} \to p, \qquad 0 < p < 1,$$

then, for fixed K and x,

$$P(X = x \mid M, N, K) \longrightarrow \binom{K}{x} p^x (1-p)^{K-x}.$$

Solution: For a fixed integer r and $n \to \infty$, Stirling's approximation gives

$$n! \approx \sqrt{2\pi n}\left(\frac{n}{e}\right)^n,$$

and consequently

$$\binom{n}{r} \sim \frac{n^r}{r!}.$$

Since

$$\frac{M}{N} \to p,$$

we have

$$M \sim pN$$

and

$$N - M \sim (1-p)N.$$

For fixed x and K,

$$\binom{M}{x} \sim \frac{M^x}{x!},$$

$$\binom{N-M}{K-x} \sim \frac{(N-M)^{K-x}}{(K-x)!},$$

and

$$\binom{N}{K} \sim \frac{N^K}{K!}.$$

Therefore,

$$P(X = x \mid M, N, K) = \frac{\binom{M}{x}\binom{N-M}{K-x}}{\binom{N}{K}} \sim \frac{\frac{M^x}{x!}\frac{(N-M)^{K-x}}{(K-x)!}}{\frac{N^K}{K!}}.$$

Hence,

$$P(X = x \mid M, N, K) \sim \frac{K!}{x!(K-x)!}\left(\frac{M}{N}\right)^x \left(\frac{N-M}{N}\right)^{K-x}.$$

$$\frac{N-M}{N} = 1 - \frac{M}{N} \to 1 - p.$$

Thus,

$$\boxed{P(X = x \mid M, N, K) \longrightarrow \binom{K}{x} p^x (1-p)^{K-x}.}$$

Therefore, for fixed sample size K, the hypergeometric distribution approaches the Binomial(K, p) distribution as the population size becomes large with the population proportion of successes approaching p.

3. Let X be a random variable with moment-generating function

$$M_X(t) = E(e^{tX}), \qquad -h < t < h.$$

Prove that

$$P(X \geq a) \leq e^{-at} M_X(t), \qquad 0 < t < h.$$

Solution: For $t > 0$, since the exponential function is increasing, the event

$$\{X \geq a\}$$

is the same as the event

$$\{e^{tX} \geq e^{ta}\}.$$

Applying Markov's inequality to the nonnegative random variable e^{tX},

$$\begin{aligned} P(X \geq a) &= P(e^{tX} \geq e^{ta}) \\ &\leq \frac{E(e^{tX})}{e^{ta}} \\ &= e^{-at} M_X(t). \end{aligned}$$

Teaching-team notes. An alternate proof appears below.
For $t > 0$, on the event $\{X \geq a\}$,

$$tX \geq ta.$$

Since the exponential function is increasing,

$$e^{tX} \geq e^{ta}.$$

Let

$$I_{\{X \geq a\}} = \begin{cases} 1, & X \geq a, \\ 0, & X < a. \end{cases}$$

If $X \geq a$, then

$$1 \leq e^{-at} e^{tX}$$

while if $X < a$, then

$$0 \leq e^{-at} e^{tX}$$

Therefore,

$$I_{\{X \geq a\}} \leq e^{-at} e^{tX}.$$

Taking expectations on both sides,

$$\begin{aligned} P(X \geq a) &= E\left[I_{\{X \geq a\}}\right] \\ &\leq e^{-at} E(e^{tX}) \\ &= e^{-at} M_X(t). \end{aligned}$$

Hence,

4. $X \sim \text{Binomial}(n, p)$. Our goal is to determine the value of k for which $P(X = k)$, is maximized.

   (a) Show that

$$P(X = k+1) = \frac{p}{1-p}\frac{n-k}{k+1}P(X = k), \qquad k = 0, 1, \ldots, n-1.$$

   Solution: For a binomial random variable,

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}.$$

   Therefore,

$$P(X = k+1) = \binom{n}{k+1} p^{k+1} (1-p)^{n-k-1}.$$

   Taking the ratio,

$$\begin{aligned} \frac{P(X = k+1)}{P(X = k)} &= \frac{\binom{n}{k+1}}{\binom{n}{k}} \frac{p}{1-p} \\ &= \frac{n-k}{k+1}\frac{p}{1-p}. \end{aligned}$$

$$\boxed{P(X = k+1) = \frac{p}{1-p}\frac{n-k}{k+1}P(X = k).}$$

   (b) Assume $0 < p < 1$, and n is sufficiently large so that

$$\frac{1}{n+1} < p < \frac{n}{n+1}.$$

   Show that, as k goes from 0 to n, the sequence $P(X = k)$ first increases and then decreases, except possibly for a plateau at the maximum. Determine the value of k for which $P(X = k)$ is maximum. Discuss when two values tie.

   ~~Show that as k goes from 0 to n, the sequence $P(X = k)$ first increases and then decreases. Determine the value of k for which $P(X = k)$ is maximum. Discuss when two values tie.~~

   Solution: From part (a),

$$\frac{P(X = k+1)}{P(X = k)} = \frac{p}{1-p}\frac{n-k}{k+1}.$$

   The sequence is increasing when

$$\frac{P(X = k+1)}{P(X = k)} > 1.$$

   Since $0 < p < 1$, this is equivalent to

$$p(n-k) > (1-p)(k+1).$$

   Simplifying,

$$np - pk > k + 1 - pk - p,$$

   so

$$(n+1)p > k + 1.$$

   Thus,

$$P(X = k+1) > P(X = k) \quad \Longleftrightarrow \quad k + 1 < (n+1)p.$$

Similarly,

$$P(X = k+1) < P(X = k) \quad \Longleftrightarrow \quad k+1 > (n+1)p.$$

Therefore, the probabilities first increase and then decrease, except possibly for the plateau.
Let

$$m = \lfloor (n+1)p \rfloor .$$

If $(n+1)p$ is not an integer, then

$$\boxed{m = \lfloor (n+1)p \rfloor}$$

is the unique value of k at which the maximum occurs.
If $(n+1)p$ is an integer, say $(n+1)p = m$, then

$$P(X = m) = P(X = m-1),$$

so there are two modes, $m-1$ and m.
In either case, the largest integer less than or equal to $(n+1)p$ is a value of k at which the maximum is attained.
Hence,

$$\boxed{\begin{cases} \lfloor (n+1)p \rfloor , & \text{if } (n+1)p \notin \mathbb{Z}, \\ (n+1)p - 1 \text{ and } (n+1)p, & \text{if } (n+1)p \in \mathbb{Z}. \end{cases}}$$

5. A stick of length 1 metre is broken at two points chosen independently and uniformly at random along its length. Determine the expected lengths of the largest and smallest pieces.

Teaching-team notes. Let the two breaking points, in increasing order, be $U_{(1)}$ and $U_{(2)}$. Express the three piece lengths in terms of these variables. Reason on the constraints in two dimensions.

Solution: Let the two breaking points, in increasing order, be $U_{(1)}$ and $U_{(2)}$. The three piece lengths are

$$L_1 = U_{(1)}, \qquad L_2 = U_{(2)} - U_{(1)}, \qquad L_3 = 1 - U_{(2)}.$$

Consider the unit square corresponding to the two ordered breaking points $(U_{(1)}, U_{(2)})$. Since $U_{(1)} < U_{(2)}$, the possible values lie in the triangle above the line

$$U_{(2)} = U_{(1)},$$

which has area $1/2$. The two problems of expected lengths will require how this feasible area is subdivided.

Smallest piece

Let S denote the smallest of the three pieces. We cannot have all 3 pieces greater than $1/3$, so we consider $0 < x < 1/3$, and look for

$$P(S > x) = P(L_1 > x, L_2 > x, L_3 > x).$$

We sketch three relevant boundaries to find the allowed region. Pick an $x < 1/3$.

- Now $S > x$ means that the first boundary has $U_{(1)} > x$. Think of this as a vertical line.

- Another boundary comes from the horizontal line $U_{(2)} < 1 - x$ since $L_3 = 1 - U_{(2)}$ and $L_3 > x$.
- The third boundary is the inclined line $U_{(2)} = U_{(1)} + x$ with "y-intercept" as the value x. (remember, the axes are represented by $(U_{(1)}, U_{(2)})$).

These three boundaries carve out a small triangle inside the allowed triangle of area $1/2$. Checking out the coordinates of this triangle we get a right triangle with two perpendicular sides with length $(1 - x) - 2x = 1 - 3x$. Hence this area $(1 - 3x)^2/2$, while the entire ordered region has area $1/2$. Hence,

$$P(S > x) = (1 - 3x)^2, \qquad 0 < x < \frac{1}{3}.$$

The cumulative distribution function is therefore

$$F_S(x) = P(S \leq x) = 1 - P(S > x) = 1 - (1 - 3x)^2.$$

Thus, for $0 < x < 1/3$,

$$F_S(x) = 6x - 9x^2.$$

Differentiating,

$$f_S(x) = 6 - 18x, \qquad 0 < x < \frac{1}{3}.$$

Therefore,

$$\begin{aligned} E(S) &= \int_0^{1/3} x f_S(x)\, dx \\ &= \int_0^{1/3} x(6 - 18x)\, dx \\ &= \int_0^{1/3} (6x - 18x^2)\, dx \\ &= \left[3x^2 - 6x^3\right]_0^{1/3} \\ &= \frac{1}{9}. \end{aligned}$$

Largest piece

Now let M denote the largest piece. We will look for $P(M \leq x)$.

1. For $0 < x < 1$,
$$P(M \leq x) = 0, \qquad x < \frac{1}{3}.$$
2. For $1/3 \leq x \leq 1/2$, the three boundary conditions result in
   - $U_{(1)} \leq x$
   - $U_{(2)} \geq 1 - x$
   - $U_{(2)} \leq U_{(1)} + x$

   This results in a triangle with area $0.5(3x - 1)^2$. Since the entire ordered region has area $1/2$,
$$F_M(x) = P(M \leq x) = \frac{\frac{1}{2}(3x - 1)^2}{\frac{1}{2}} = (3x - 1)^2, \qquad \frac{1}{3} \leq x \leq \frac{1}{2}.$$

3. For $\frac{1}{2} \le x < 1$, two different pieces cannot both exceed x. Indeed, if for example

$$L_1 > x \qquad \text{and} \qquad L_2 > x,$$

then

$$L_1 + L_2 > 2x \ge 1,$$

which is impossible since

$$L_1 + L_2 + L_3 = 1$$

and $L_3 \ge 0$.

Thus, for $x \ge 1/2$, the three events

$$\{L_1 > x\}, \qquad \{L_2 > x\}, \qquad \{L_3 > x\}$$

are mutually disjoint.

Since

$$M = \max(L_1, L_2, L_3),$$

we have

$$\{M > x\} = \{L_1 > x\} \cup \{L_2 > x\} \cup \{L_3 > x\}.$$

Therefore,

$$\begin{aligned} P(M > x) &= P(L_1 > x) + P(L_2 > x) + P(L_3 > x) \\ &= 3(1-x)^2. \end{aligned}$$

In the last step, we have determined the individual densities again by considering the unit square and figuring out the boundaries as in the previous cases. Specifically $P(L_1 \ge x)$ will simply create a corner triangle of side lengths $(1-x)$, and thus the probability is $(1-x)^2$. By symmetry, $P(L_3 \ge x)$ is the same. For the middle piece, $P(L_2 > w) = P((U_{(2)} - U_{(1)}) > w)$, and again using the geometry of the unit square, we have

$$P(L_2 > w) = (1-w)^2, \qquad 0 < w < 1.$$

Returning to the largest piece M, we thus have

$$F_M(x) = P(M \le x) = 1 - 3(1-x)^2, \qquad \frac{1}{2} \le x < 1.$$

Differentiating the cumulative distribution functions for the three parts, we obtain

$$f_M(x) = \begin{cases} 18x - 6, & \frac{1}{3} < x < \frac{1}{2}, \\ 6(1-x), & \frac{1}{2} < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Therefore,

$$\begin{aligned} E(M) &= \int_{1/3}^{1/2} x(18x-6)\,dx + \int_{1/2}^{1} x\,6(1-x)\,dx \\ &= \left[6x^3 - 3x^2\right]_{1/3}^{1/2} + \left[3x^2 - 2x^3\right]_{1/2}^{1} \\ &= \frac{1}{9} + \frac{1}{2} \\ &= \frac{11}{18}. \end{aligned}$$

Thus,

$$\boxed{E(M) = \frac{11}{18}}.$$

6. Let $U \sim \text{Uniform}(0,1)$ and define

$$X = U(1-U).$$

Find the cumulative distribution function and probability density function of X.

Solution: Since

$$X = U(1-U) = \frac{1}{4} - \left(U - \frac{1}{2}\right)^2,$$

the range of X is

$$0 \le x \le \frac{1}{4}.$$

For $0 < x < 1/4$,

$$\begin{aligned} F_X(x) &= P(U(1-U) \le x) \\ &= P\left(U^2 - U + x \ge 0\right). \end{aligned}$$

The roots of

$$u^2 - u + x = 0$$

are

$$u = \frac{1 \pm \sqrt{1-4x}}{2}.$$

We will use the roots to find the region where the quadratic is non-negative and also apply the constraint on $U \in [0,1]$.

Therefore,

$$U(1-U) \le x$$

when

$$0 \le U \le \frac{1-\sqrt{1-4x}}{2}$$

or

$$\frac{1+\sqrt{1-4x}}{2} \le U \le 1.$$

Since U is uniform on $(0,1)$, we simply have to add up the lengths:

$$\begin{aligned} F_X(x) &= \frac{1-\sqrt{1-4x}}{2} + (1 - \frac{1+\sqrt{1-4x}}{2}) \\ &= 1 - \sqrt{1-4x}. \end{aligned}$$

Hence,

$$F_X(x) = \begin{cases} 0, & x < 0, \\ 1 - \sqrt{1-4x}, & 0 \le x \le \frac{1}{4}, \\ 1, & x > \frac{1}{4}. \end{cases}$$

Differentiating,

$$f_X(x) = \frac{2}{\sqrt{1-4x}}, \qquad 0 < x < \frac{1}{4}.$$

Thus,

$$\boxed{F_X(x) = \begin{cases} 0, & x < 0, \\ 1 - \sqrt{1-4x}, & 0 \le x \le \frac{1}{4}, \\ 1, & x > \frac{1}{4}, \end{cases}}$$

and

$$\boxed{f_X(x) = \begin{cases} \dfrac{2}{\sqrt{1-4x}}, & 0 < x < \dfrac{1}{4}, \\ 0, & \text{otherwise.} \end{cases}}$$

7. Let $X \sim \text{Binomial}(n, p)$. Define

$$Y = \frac{1}{X+1}.$$

(a) Show that

$$E\left(\frac{1}{X+1}\right) = \frac{1-(1-p)^{n+1}}{(n+1)p}.$$

Solution: By the binomial probability mass function,

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}.$$

Therefore,

$$E\left(\frac{1}{X+1}\right) = \sum_{k=0}^{n} \frac{1}{k+1} \binom{n}{k} p^k (1-p)^{n-k}.$$

Using

$$\frac{1}{k+1}\binom{n}{k} = \frac{1}{n+1}\binom{n+1}{k+1},$$

we obtain

$$E\left(\frac{1}{X+1}\right) = \frac{1}{n+1} \sum_{k=0}^{n} \binom{n+1}{k+1} p^k (1-p)^{n-k}.$$

Let $j = k+1$. Then

$$E\left(\frac{1}{X+1}\right) = \frac{1}{(n+1)p} \sum_{j=1}^{n+1} \binom{n+1}{j} p^j (1-p)^{n+1-j}.$$

By the binomial theorem,

$$\sum_{j=0}^{n+1} \binom{n+1}{j} p^j (1-p)^{n+1-j} = 1.$$

Removing the $j = 0$ term gives

$$\sum_{j=1}^{n+1} \binom{n+1}{j} p^j (1-p)^{n+1-j} = 1 - (1-p)^{n+1}.$$

Therefore,

$$\boxed{E\left(\frac{1}{X+1}\right) = \frac{1-(1-p)^{n+1}}{(n+1)p}.}$$

(b) Deduce a closed-form expression for

$$E\left(\frac{X}{X+1}\right).$$

Solution: Observe that

$$\frac{X}{X+1} = 1 - \frac{1}{X+1}.$$

Therefore,

$$\begin{aligned} E\left(\frac{X}{X+1}\right) &= 1 - E\left(\frac{1}{X+1}\right) \\ &= 1 - \frac{1-(1-p)^{n+1}}{(n+1)p}. \end{aligned}$$

Thus,

$$\boxed{E\left(\frac{X}{X+1}\right) = 1 - \frac{1-(1-p)^{n+1}}{(n+1)p}.}$$